

3. (2 points)

Given the following information:

<u>Cumulative Reported Loss (\$000)</u>		
Accident	12	24
<u>Year</u>	<u>Months</u>	<u>Months</u>
2011	1,000	1,950
2012	1,200	2,650
2013	1,800	3,500
2014	1,500	3,300
2015	2,000	4,100
2016	2,400	

- $\text{Var}(C_{i,k+1}|C_{i1},\dots,C_{ik})$ is constant for all i where $C_{i,k}$ is the accumulated total claims amount of accident year i up to development year k .

a. (1 point)

Calculate the age-to-age factor for 12 to 24 months using the most appropriate formula according to Mack.

b. (1 point)

Plot the residuals against the cumulative reported loss as of 12 months and interpret the results.

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QUESTION 3

TOTAL POINT VALUE: 2

LEARNING OBJECTIVE(S): A2

SAMPLE ANSWERS

Part a: 1 point

Sample 1

As variance is constant

$$f_{12-24} = \frac{(1000 * 1950 + 1200 * 2650 + \dots + 2000 * 4100)}{(1000^2 + 1200^2 + \dots + 2000^2)}$$

$$f_{12-24} = 2.06$$

Sample 2

Var proportional to 1 -> Cik^2 weighted factor

$$f_{12-24} = \frac{(1000^2 * (\frac{1950}{1000}) + 1200^2 * (\frac{2650}{1200}) + \dots + 2000^2 * (\frac{4100}{2000}))}{(1000^2 + 1200^2 + \dots + 2000^2)}$$

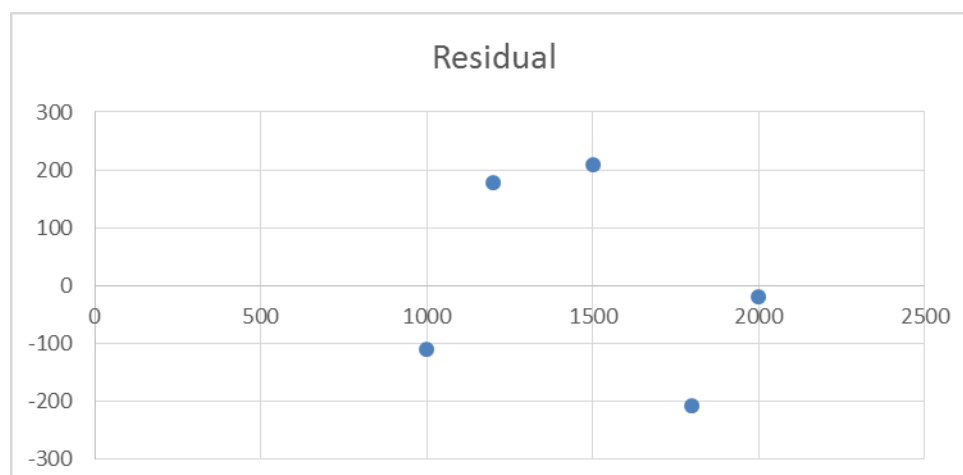
$$f_{12-24} = 2.060$$

Part b: 1 point

Sample 1

Residuals = $C_{i,k+1} - C_{i,k} * f_{k0}$

Accident Year	Loss at 12 months	Expected Loss at 24 months	Actual Loss at 24 months	Residual
2011	1000	$1000 * 2.060 = 2060$	1950	$1950 - 2060 = -110$
2012	1200	2472	2650	178
2013	1800	3709	3500	-209
2014	1500	3091	3300	209
2015	2000	4121	4100	-21



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Sample Interpretation 1

Residuals are randomly scattered around zero so the variance assumption is reasonable.

Sample Interpretation 2

Since the residuals are randomly scattered around zero with no trend or change in magnitude, this variance assumption seems reasonable.

EXAMINER'S REPORT

Candidates were expected to understand the relationship between (a) variance assumptions which diverged from the original chain ladder assumptions given in Mack (1994) and (b) methods of calculating loss development factors given the alternate variance assumptions.

Candidates were also expected to understand how to calculate residuals given alternate variance assumptions and how to interpret the results of those residual calculations in order to determine how well the given model fit the data.

Part a

Candidates were expected to understand that the variance assumption stated in the problem (constant variance) differed from the original Mack (1994) chain ladder assumptions. Candidates were expected to apply the correct formula to calculate the loss development factor from 12-24 months, given the alternate variance assumption.

A common error was applying the incorrect formula given the variance assumption stated in the problem

Part b

Candidates were expected to calculate the residuals by accident year given the variance assumption stated in the problem. Candidates were also expected to understand how to plot the residuals graphically and interpret the results to determine if the stated variance assumption was reasonable.

Common errors include:

- Using an incorrect formula to calculate residuals given the development factor calculation in Part a.
- Incorrect labeling on the x-axis of the graph (years, cumulative losses at 24 months, etc.)
- Not giving an interpretation for the residuals – some candidates only stated what the residual plot looked like without giving an interpretation of the results